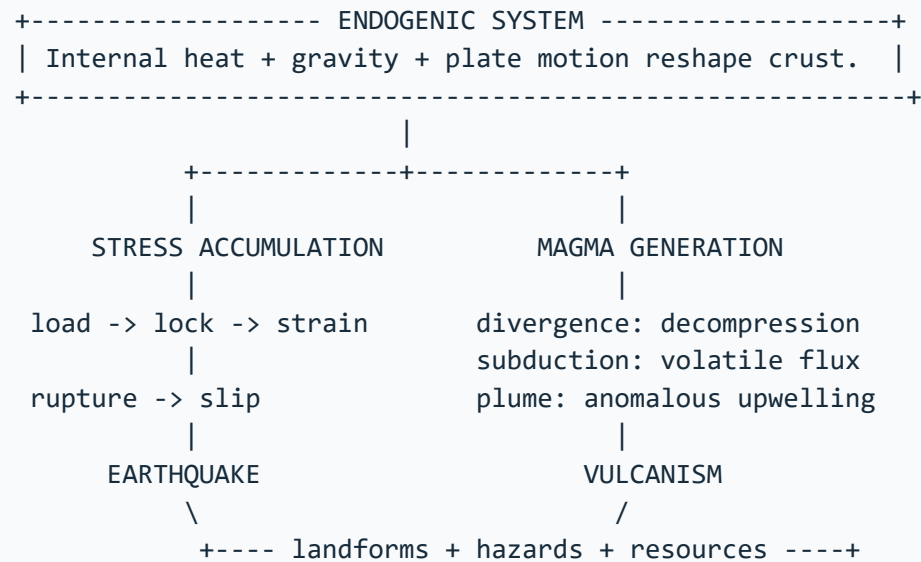


ASCII MASTER FLOW – PANEL 1/12: Endogenic system: plate motion, strain release and magma generation



FAULT PAIRS

tension -> normal | compression -> reverse/thrust | shear -> strike-slip

Plate motion loads a region; sudden fault rupture is the immediate quake source.

MUST REMEMBER: Vulcanism and seismicity follow magma generation, ascent, eruption style, fault rupture, wave propagation and site response; India requires Himalayan collision, Kachchh intraplate strain, Peninsular reservoirs/faults, Andaman subduction and Barren Island anchors.

MUST REMEMBER: Vulcanism and seismicity follow magma generation, ascent, eruption style, fault rupture, wave propagation and site response; India requires Himalayan collision, Kachchh intraplate strain, Peninsular reservoirs/faults, Andaman subduction and Barren Island anchors.

ASCII MASTER FLOW – PANEL 2/12: Earthquake source geometry, depth and seismogram logic

SURFACE
station A ----- epicentre ----- station B
|
|
focus / hypocentre
rupture begins on fault plane

SEISMOGRAM ARRIVAL ORDER

P first -> S second -> Love and Rayleigh later

|
P-S interval at one station gives source distance

|
three or more distance circles estimate the epicentre

DEPTH CLASS

shallow: 0-70 km | intermediate: 70-300 km | deep: 300-700 km

SEQUENCE FIREWALL

foreshock, mainshock and aftershock are relational labels;
a swarm has no single clearly dominant event.

Epicentre is a vertical projection, not necessarily the worst-damaged place.

ASCII MASTER FLOW – PANEL 3/12: Seismic waves, shadow zones, magnitude and intensity

WAVE	MOTION	MEDIUM	MAIN CLUE
P	compression	solid, liquid	core refraction
S	transverse shear	solid only	liquid outer core
Love	horizontal shear	surface	lateral shaking
Rayleigh	rolling ellipse	surface	mixed motion

P shadow about 103-142 deg | direct S shadow beyond about 103 deg

MAGNITUDE: one source measure; Mw uses seismic moment

moment = rigidity x fault area x average slip

+1 local magnitude gives about 10x amplitude and 31.6x energy

INTENSITY: place-specific observed effects; many values for one event

source x path x site x exposure x vulnerability / capacity -> damage

ASCII MASTER FLOW – PANEL 4/12: Global seismic belts, cascading hazards and tsunami sequence

GLOBAL EARTHQUAKE DISTRIBUTION

```
+-----+-----+-----+
| Circum-Pacific      | Mediterranean-    | Ridges/transforms  |
| trench + slab + arc | Himalayan collision| shallow events      |
| megathrust + tsunami| complex convergence| basaltic ridges     |
+-----+-----+-----+
Intraplate events reactivate inherited structures; stable is not aseismic.
```

CASCADE

```
shaking -> amplification / liquefaction / lateral spreading
         -> landslide / fire / lifeline failure / dam distress
```

TSUNAMI

```
submarine megathrust -> vertical sea-floor displacement
-> long low deep-ocean wave -> coastal shoaling
-> run-up + inundation + return flow
```

First wave need not be largest; withdrawal need not occur before every event.
Risk varies with bathymetry, bay shape, shelf, exposure and evacuation.

ASCII MASTER FLOW – PANEL 5/12: Volcano anatomy, magma controls, landforms and eruption styles

```
      ash plume
        /\
       /  \ crater
lava  --    -- lava
      |vent|
dyke  ----|   |---- sill
      |conduit
      magma chamber
```

Magma below surface | lava after eruption

Silica up -> viscosity up -> degassing harder -> explosivity may rise

Temperature up -> viscosity generally down

Retained gas + decompression -> bubbles -> fragmentation

FORMS: batholith, laccolith, lopolith, phacolith, dyke, sill

Dyke cuts layers; sill follows layers.

STYLE: fissure/Icelandic -> Hawaiian -> Strombolian -> Vulcanian -> Plinian

FORM: shield | composite | cinder cone | dome | fissure plateau

STATUS: active | dormant | extinct; status is not a compulsory life cycle.

ASCII MASTER FLOW – PANEL 6/12: Volcanic settings, global belts, products and environmental effects

+-----+	+-----+	+-----+
SETTING	MELTING MECHANISM	EXPRESSION
+-----+	+-----+	+-----+
Divergent	decompression	ridge, rift, basalt
Subduction	slab volatile flux	trench-slab-arc system
Plume	mantle upwelling	flood basalt, hot spot
+-----+	+-----+	+-----+

Ring of Fire = trench + descending slab + earthquakes + volcanic arc
Plume model: broad head may form flood basalt; tail may form age chain.
Deccan Trap is a fissure-fed flood-basalt plateau, not a giant cone.

HAZARDS: lava | ash | density currents | lahars | gases
CLIMATE: stratospheric sulphate aerosol can cause temporary cooling
BENEFITS: new crust/land | soil parent | geothermal energy | minerals

Impact converges through wind, relief, drainage, exposure and preparedness.

ASCII MASTER FLOW – PANEL 7/12: India seismic geography, zonation, institutions and resilience stack

INDIA REGIONAL HIERARCHY

Himalaya : collision, thrusts, slopes, dense corridors
North-East : convergence, strike-slip complexity
Andaman : subduction, tsunami, Barren Island volcanism
Kachchh : inherited rift and fault reactivation
Peninsula : relatively stable; Koyna and Latur show residual risk
Indo-Gangetic : alluvium, amplification and high exposure

|

MACROZONATION: broad design hazard, not prediction

MICROZONATION: geology + soil + motion + water + liquefaction + GIS

|

NCS/MoES -> monitoring | GSI -> geology | BIS -> standards

NDMA/NIDM -> guidance | INCOIS -> tsunami warning

States/ULBs -> approvals + inspection + enforcement

|

site -> regular load path -> ductility -> connections

-> non-structural safety -> quality control -> retrofit -> drills

Early warning detects rupture after it starts; it is not prediction.

ASCII MASTER FLOW – PANEL 8/12: Hazard-risk-response and Mains answer spine for seismic India

QUESTION: Why can similar earthquakes produce unequal disasters?

HAZARD: source size + depth + rupture geometry + path

SITE: basin sediment + groundwater + slope + liquefaction potential

EXPOSURE: people + buildings + lifelines + time of occurrence

VULNERABILITY: weak design + irregular form + poor maintenance

CAPACITY: codes + enforcement + retrofit + warning + response

RISK VERDICT: hazard becomes disaster through social and spatial filters

15-MARK INDIA SPINE

tectonic regions -> site modifiers -> exposure/vulnerability -> cases

-> zonation and microzonation -> construction and governance -> conclusion

TRAPS: focus is not epicentre | magnitude is not intensity

tsunami is not a tide | caldera is not an ordinary crater

zone map is not prediction | code publication is not enforcement.

ASCII MASTER FLOW – PANEL 9/12: Evidence ladder and confidence control

EVIDENCE LADDER

1. Convert relief, water, climate and resources into a dated historical mechanism.
2. Add technology, labour, institutions and power before claiming an outcome.
3. State each proxy's catchment, chronology, preservation and alternative causes.

VERDICT: Geography conditions choices; it does not dictate one inevitable history.

ASCII MASTER FLOW – PANEL 10/12: Examiner traps and contested boundaries

CLOSE DISTINCTIONS

1. Mountains are barriers and corridors; rivers are routes and changing hazards.
2. A palaeochannel is not a named textual river; correlation is not causation.
3. Resource maps are not political maps; modern borders are not ancient regions.

VERDICT: Date the landscape, match the scale and preserve regional variation.

CLOSE DISTINCTION: Magnitude measures source size whereas intensity records effects; focus differs from epicentre, P differs from S and surface waves, and a tsunami requires water-column displacement rather than every submarine earthquake.

ASCII MASTER FLOW – PANEL 11/12: Integrated answer spine and qualified conclusion

ANSWER SPINE

1. For 2023 GS-I Q1, map mountain, plain, plateau, river, resource and coast.
2. Write factor -> mechanism -> named evidence -> outcome -> mediation -> limit.
3. Compare prehistoric, Harappan, Vedic and early historic uses of landscapes.

VERDICT: Societies selected, connected and transformed geographic possibilities.

EVIDENCE LIMIT: Hazard is not disaster: exposure, vulnerability, construction and preparedness mediate loss; seismic zonation is probabilistic and revised through evidence, while prediction claims must not be confused with monitoring or early warning.

EVIDENCE LADDER

1. = from source book = inference / standard knowledge = current affairs.
 2. Extinct volcano Geological evidence indicates eruption is no longer...
 3. geophysical evidence; a long-quiescent volcano may still be dormant.
- VERDICT: Source type controls the strength and limits of the historical claim.